

Economic Growth, the Financial System, and Business Cycles

ECON 101 – Principles of Macroeconomics

Muhebullah Karimzada

Spring 2026

- 6.1 Long-Run Economic Growth
- 6.2 Saving, Investment, and the Financial System
- 6.3 The Business Cycle

6.1 Learning Objective

Goal

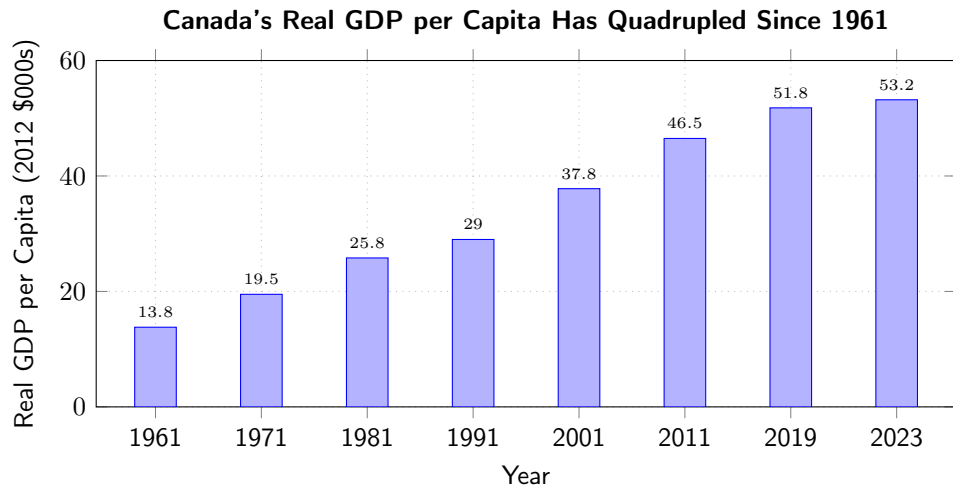
Explain how long-run economic growth is measured and what determines it.

- **Long-run economic growth:** the process by which rising **productivity** increases the **average standard of living**.
- This differs from short-run fluctuations (**business cycles**):
 - **Expansion:** real GDP rising.
 - **Recession:** real GDP falling.
- The most common measure of the standard of living:
 - **Real GDP per capita** = $\text{real GDP} \div \text{population}$.

Real GDP per Capita: A Key Indicator

- **Real GDP per capita** tells us:
 - How much goods and services, on average, each person can consume.
 - Adjusted for changes in the price level (**real**, not nominal).
- Countries with higher real GDP per capita typically have:
 - better health, education, housing, and infrastructure.
- **Canadian context:** In 2023, Canada's real GDP per capita was approximately **\$53,000** (2012 constant dollars), roughly **four times** the 1961 level of $\approx$ \$13,800.
 - **Source:** Statistics Canada, Table 36-10-0222-01

Figure 6.1 – Real GDP per Capita in Canada, Selected Years



Source: Statistics Canada, Table 36-10-0222-01 (2012 constant dollars).

Interpretation of Figure 6.1

- Measured in 2012 dollars, real GDP per capita has risen **nearly four-fold** since 1961.
- The average Canadian in 2023 commands **almost four times as many** goods and services as the average Canadian in 1961.
- This long-run rise is what we call **economic growth**.
- **Notable interruptions:**
 - Early 1980s recession (high interest rates, oil price shock).
 - Early 1990s recession (tight monetary policy, housing bust).
 - 2008–2009 Global Financial Crisis.
 - **2020 COVID-19 pandemic:** real GDP per capita fell $\approx 13\%$ in just 2 quarters — the sharpest decline on record.

Calculating Growth Rates

- Growth rate of an economic variable from year $t - 1$ to year t :

$$\text{Growth rate}_t = \frac{\text{Value}_t - \text{Value}_{t-1}}{\text{Value}_{t-1}} \times 100.$$

- For a few years, compute growth each year and take the average.
- **Canadian example:**
 - Real GDP per capita in 2022: \$53,100.
 - Real GDP per capita in 2023: \$53,200.
 - Growth rate $\approx \frac{53,200 - 53,100}{53,100} \times 100 \approx \mathbf{0.19\%}$.
 - **Canada's 2023 growth was very weak — immigration drove population faster than real output.**

The Rule of 70

- A handy shortcut: the **Rule of 70**.
- If a variable grows at rate $g\%$ per year, the number of years to **double** is approximately:

$$\text{Years to double} \approx \frac{70}{g}.$$

- **Canadian examples:**
 - Canada's historical average growth of real GDP per capita $\approx 2\%$:
 - Years to double $\approx 70/2 = 35$ years.
 - At only 1% growth: $70/1 = 70$ years — twice as long!
- Small differences in growth rates **compound** dramatically over decades.

Rule of 70: Country Comparison

Country A — 2% growth

- Years to double: $70/2 = 35$
- Income after 35 years: **2x**
- Income after 70 years: **4x**

Country B — 4% growth

- Years to double: $70/4 = 17.5$
- Income after 35 years: **4x**
- Income after 70 years: **16x**

- **Real example:** South Korea's economy grew $\approx 7\%$ per year from 1960 to 2000, doubling every 10 years — raising per capita income from $\approx \$1,000$ to $\approx \$20,000$ (2000 US\$).
- Canada's steadier $\approx 2\%$ average represents **sustained prosperity without boom-and-bust volatility**.

What Determines Long-Run Growth?

- Long-run increases in real GDP per capita depend on **labour productivity**:

$$\text{Labour productivity} = \frac{\text{Real GDP}}{\text{Number of workers (or hours worked)}}.$$

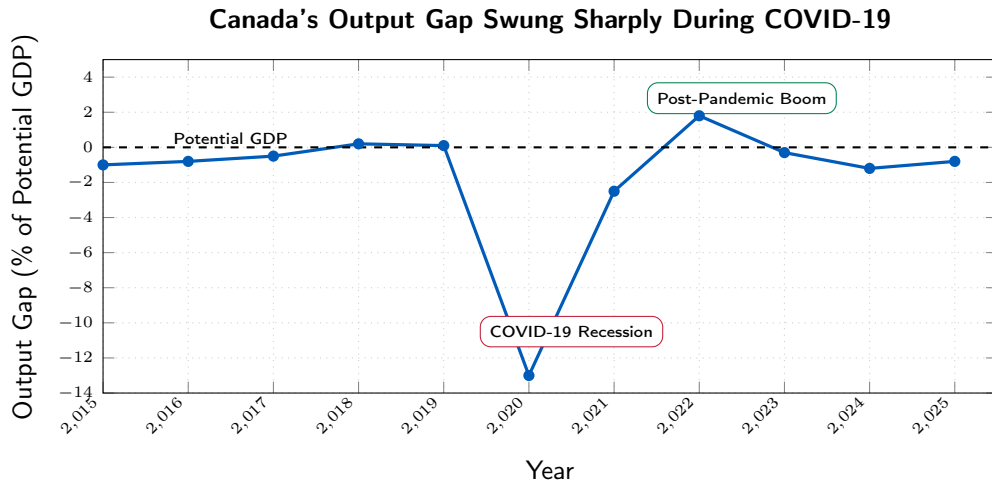
- The average Canadian produces much more per hour than in 1961.
- Two key drivers:
 1. **Increases in capital per hour worked**
 2. **Technological change**
- **Canadian data (Statistics Canada):** Labour productivity grew an average of **1.4%** per year from 2000–2023 — slower than the G7 average, raising policy concern.

- **Physical capital:** machines, computers, tools, buildings, vehicles.
 - More capital per worker $\Rightarrow$ more output per hour.
- **Human capital:**
 - Accumulated knowledge and skills from education, training, experience.
- **Canadian example:**
 - A Canadian oil sands worker equipped with modern extraction technology and earth-moving equipment produces orders of magnitude more output per day than a 1961 worker with manual tools.
 - BC's tech sector: engineers with university degrees and high-powered workstations represent both physical and human capital investment.

- **Technological change:** improvements in methods of combining inputs into outputs.
- Examples:
 - New production processes (AI-assisted manufacturing).
 - Better management (lean production, just-in-time inventory).
 - New products (mRNA vaccines, electric vehicles).
- Workers produce more **with the same inputs** — an outward shift of the production frontier.
- **Canadian example:** Shopify (Ottawa) enables thousands of small businesses to reach global markets with the same team size — a pure technology-driven productivity gain.
- **Entrepreneurs** pioneer these new ways of combining labour, capital, and ideas.

- **Potential GDP:** the level of real GDP attained when all firms produce at **capacity** —
 - “normal” hours, “normal” workforce, no emergency overtime.
- Also called **full-employment GDP** or **trend GDP**.
- Potential GDP grows when:
 - the labour force expands (e.g., through immigration),
 - new factories and equipment are installed,
 - technological change raises productivity.
- **Bank of Canada** estimates and monitors potential GDP as part of its quarterly Monetary Policy Report (MPR).

Figure 6.2 – Output Gap: Canada 2015–2025 (Estimated)



Source: Bank of Canada, Monetary Policy Report (estimates). Shaded area = negative output gap (recessionary).

The Output Gap

- **Output gap:** = actual GDP – potential GDP.
- When the output gap is **positive** (actual > potential):
 - economy is using resources **unsustainably** — very low unemployment, high capacity utilization, **inflationary pressure**.
 - **Canada 2022:** output gap $\approx +1.8\%$ coincided with CPI inflation peaking at **8.1%**.
- When the output gap is **negative** (actual < potential):
 - resources (labour, capital) are **underutilized**.
 - **Canada 2020:** output gap plunged to $\approx -13\%$ during COVID-19 lockdowns.
- The Bank of Canada uses the output gap to guide its monetary policy decisions (MPR, 2024).

6.2 Learning Objective

Goal

Explain how the financial system links saving and investment and why that matters for growth.

- Long-run growth requires:
 - more capital per worker, and ongoing technological improvements.
- Both require **investment**.
- Investment is financed by **saving**, channeled through the financial system.
- **Canadian saving rate** (household, % of disposable income): fell from $\approx 18\%$ in 1982 to $\approx 2\text{--}4\%$ pre-COVID, then **surged** to 28% during 2020 lockdowns (Statistics Canada).

An Overview of the Financial System

- **Financial markets:** where financial securities are bought and sold.
 - Toronto Stock Exchange (TSX), bond markets, money markets.
- **Stock:** financial security representing partial **ownership** of a firm.
 - Buying TSX-listed Shopify shares $\Rightarrow$ you own part of Shopify.
- **Bond:** financial security promising to repay a **fixed amount** of funds.
 - Government of Canada bonds, corporate bonds (e.g., TD Bank bonds).
 - A bond is effectively a **loan** from the buyer to the issuer.
- **Financial intermediaries:** firms that borrow from savers and lend to borrowers.
 - Royal Bank, TD, Scotiabank, credit unions, mutual funds, pension funds (e.g., CPPIB).

Three Key Services of the Financial System

Risk-Sharing

- Spread money across many assets.
- Reduces risk while maintaining returns.
- *E.g., RBC Select Balanced mutual fund.*

Liquidity

- Ease of converting to cash.
- Bank accounts, money market funds, publicly traded stocks.
- Allows long-term commitments with short-term access.

Information

- Prices of securities aggregate expectations.
- Guide funds toward the most productive uses.
- Reduce adverse selection and moral hazard.

The Macroeconomics of Saving and Investment

- National income identity:

$$Y = C + I + G + NX.$$

- For simplicity, assume a **closed economy** ($NX = 0$):

$$Y = C + I + G \Rightarrow I = Y - C - G.$$

- Investment equals total income not consumed or used by government.
- **Private saving:**

$$S_{\text{private}} = Y + \text{TR} - C - T.$$

- **Public saving:**

$$S_{\text{public}} = T - G - \text{TR}.$$

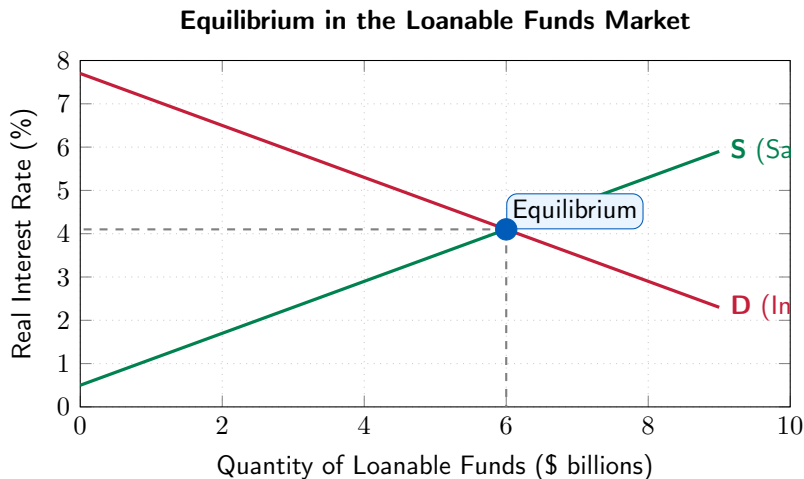
- **Total saving = Investment:**

$$S = S_{\text{private}} + S_{\text{public}} = I$$

Budget Deficits and Crowding Out

- If $S_{\text{public}} < 0$: government runs a **budget deficit**.
 - Government must borrow (sell bonds), reducing funds available for private investment.
- **Crowding out**: the reduction in private investment caused by government borrowing.
- **Canadian context**:
 - Federal government ran deficits every year 2009–2024.
 - 2020–21 deficit: **\$314 billion** (largest ever), due to COVID-19 emergency spending.
 - 2023–24 deficit: \$40 billion (Department of Finance Canada).
- In practice, crowding-out effects on interest rates are partially offset by global capital flows.

Figure 6.3 – The Market for Loanable Funds



Supply and Demand for Loanable Funds

- **Demand for loanable funds (firms):**
 - Firms borrow to finance investment projects.
 - Borrow more when real interest rate is **lower** (more projects profitable).
 - **Canada 2020**: pandemic uncertainty slashed private investment by $\approx 15\%$.
- **Supply of loanable funds (households):**
 - Households save more when real interest rate is **higher** (greater reward to saving).
 - Government saving (surplus) adds to supply; deficits reduce supply.
- **Real interest rate** adjusts to equate saving and investment.

Figure 6.4 – Increase in Demand for Loanable Funds

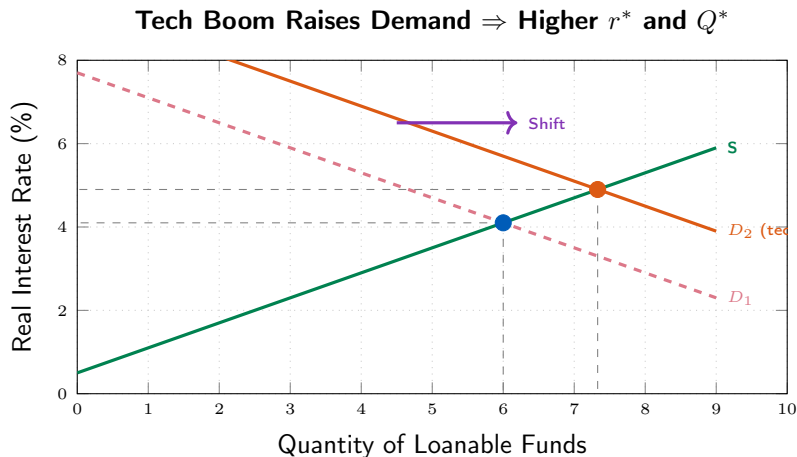
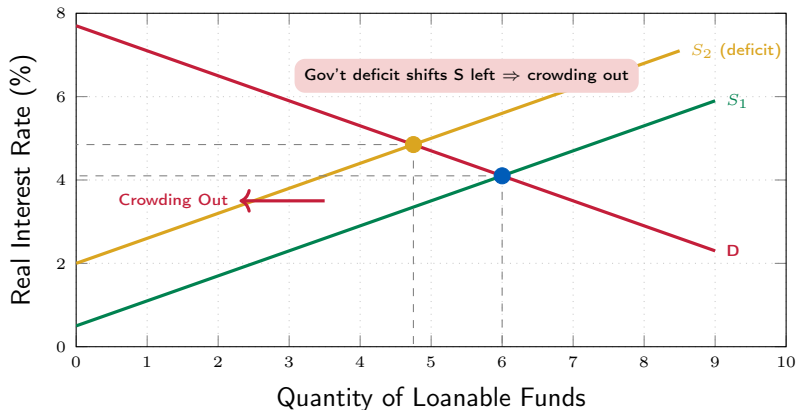


Figure 6.5 – Budget Deficit Crowds Out Investment

Government Deficit Reduces Supply $\Rightarrow$ Higher r^* , Lower Private Q^*



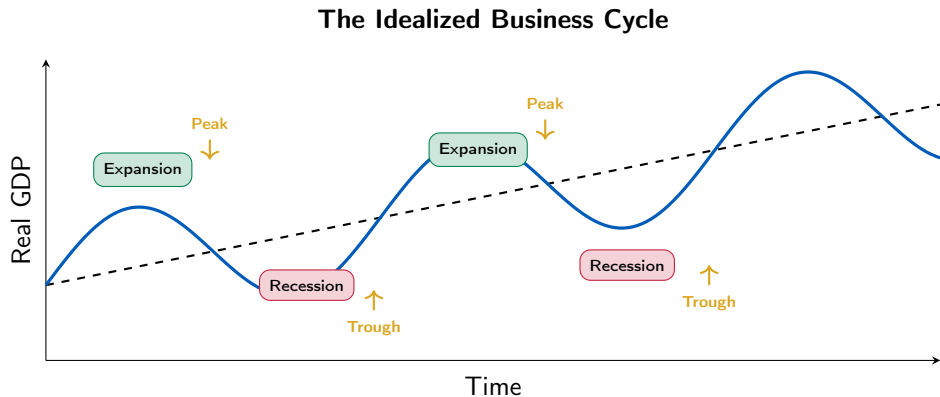
6.3 Learning Objective

Goal

Explain what happens during the business cycle and how it affects inflation and unemployment.

- Real GDP per capita has risen about **four-fold** since 1961, but not smoothly.
- Canada has experienced several recessions:
 - 1981–82, 1990–91, 2008–09, and **2020**.
- The **business cycle** describes these short-run fluctuations.

Figure 6.6 – Phases of the Business Cycle



Phases of the Business Cycle

- **Expansion:** real GDP is rising (employment grows, incomes rise).
- **Peak:** the point where real GDP stops rising and begins to fall.
- **Recession:** real GDP is falling (often defined as 2+ consecutive quarters).
- **Trough:** the point where real GDP stops falling and begins to rise.
- A new expansion follows the trough.

Canadian Recessions Since 1980

Recession	Cause	Peak Unemployment
1981–82	Volcker rate shock	13.0%
1990–91	Tight monetary policy	11.9%
2008–09	Global financial crisis	8.7%
2020	COVID-19 pandemic	14.6% (May 2020)

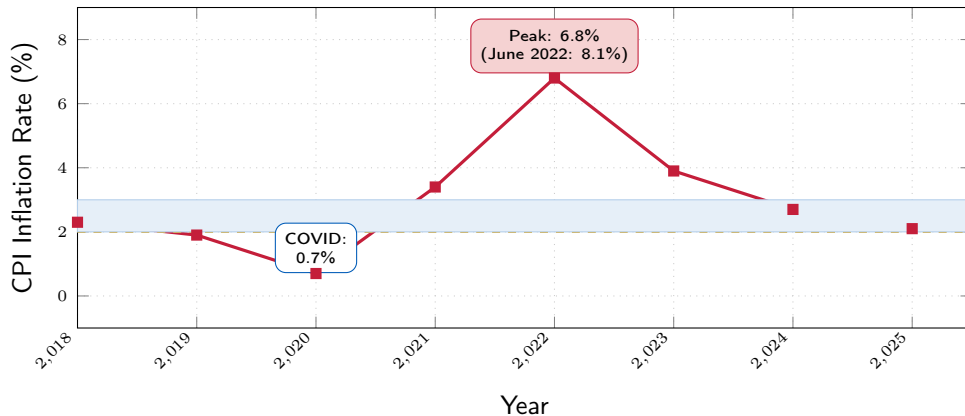
Source: Statistics Canada, Labour Force Survey.

How Does Canada Identify Recessions?

- There is **no official body** in Canada that declares recessions.
- Common rule of thumb:
 - **Two consecutive quarters** of negative real GDP growth.
- Statistics Canada (CANSIM) provides real GDP data quarterly and monthly.
- **2020 example:**
 - Q1 2020: real GDP fell -8.8% (annualized).
 - Q2 2020: real GDP fell -38.7% (annualized) — deepest quarterly drop on record.
 - Economy recovered to 2019 levels only by end of 2021.

Figure 6.7 – Inflation and the Business Cycle (Canada)

Canada CPI Inflation Fell in Recession, Surged in Recovery



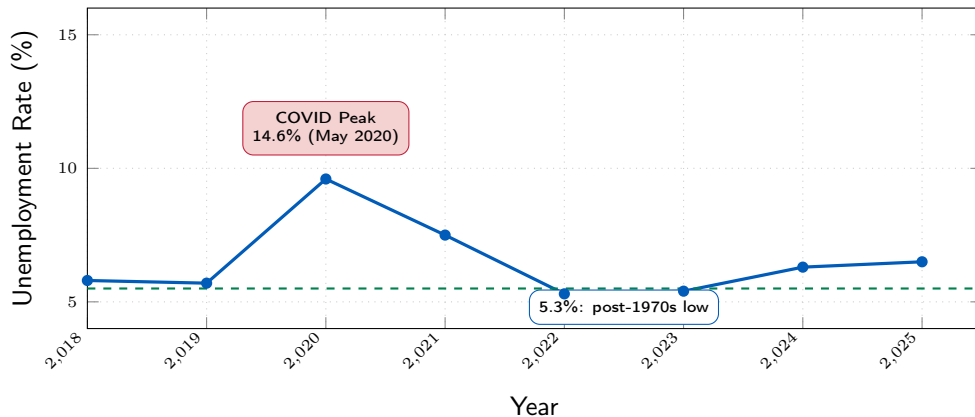
Source: Statistics Canada, Table 18-10-0004-01. 2025 is estimated. BoC target band 1–3% shaded.

The Effect of the Business Cycle on Inflation

- **During recessions:**
 - Demand falls below supply capacity.
 - Firms compete for fewer customers $\Rightarrow$ prices rise more slowly or fall.
 - **Canada 2020:** CPI inflation dropped to **0.7%**.
- **During expansions:**
 - Demand exceeds supply capacity.
 - Firms can raise prices more easily $\Rightarrow$ higher inflation.
 - **Canada 2022:** Post-COVID surge in demand + supply chain disruptions $\Rightarrow$ CPI peaked at **8.1%** in June 2022 (Statistics Canada).
- Bank of Canada responded with the **fastest rate-hike cycle in decades:** 0.25% $\rightarrow$ 5.0% (March 2022 – July 2023).

Figure 6.8 – Unemployment and the Business Cycle (Canada)

Unemployment Rose Sharply in 2020, Then Gradually Climbed Again



Source: Statistics Canada, Labour Force Survey (Table 14-10-0017-01). 2025 is estimated.

The Effect of the Business Cycle on Unemployment

- **During recessions:**
 - Firms see sales fall, reduce production, lay off workers.
 - Unemployment often **continues to rise** even after the recession begins (lagging indicator).
 - **Canada 2020:** unemployment hit **14.6%** in May 2020.
- **During expansions:**
 - Firms hire more; unemployment gradually falls.
 - **Canada 2022:** unemployment reached **5.3%**, near a 50-year low, amid very tight labour markets.
- **Canada 2023–2024:** rising immigration & slowing economy pushed unemployment back up to **6.3%** — signalling excess slack.

- **Long-run economic growth** is driven by **labour productivity**, which depends on:
 - capital per worker, human capital, and technological change.
- Canada's real GDP per capita quadrupled since 1961 — mostly due to technology and capital deepening.
- **Potential GDP** and the **output gap** reveal whether resources are over- or under-utilized.
- The **financial system** (banks, markets, intermediaries) channels saving into investment via:
 - risk-sharing, liquidity, and information.
- In a closed economy, $S = I$; government deficits can **crowd out** private investment.
- The **business cycle** — expansions, peaks, recessions, troughs — is accompanied by:
 - higher inflation in booms, lower in recessions.
 - rising unemployment in recessions, falling in expansions.
- Canada's 2020 COVID-19 recession was the sharpest in recorded history, followed by the fastest rate-hiking cycle to combat post-pandemic inflation.

Thank you!

Questions?